A twelve-generator Artin group with a CAT(0) obstruction

Abstract

We specify an Artin matrix on twelve standard generators, with all off-diagonal labels in $\{2, 3, \infty\}$, and prove that its Artin group admits no proper cocompact isometric action on a nonempty proper CAT(0) space. The proof includes the stable-length identities, a sharpened ten-strand braid estimate, the six-sector obstruction and its finite-diagram argument, the required subgroup embeddings, and the common-centralizer construction. The only Coxeter-theoretic background retained is the standard exchange/word theorem, stated explicitly in Section 6. No minimality claim is made.

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1 The construction and the theorem

This note gives a twelve-generator variation of the construction in [1]. All results used from that manuscript are proved below; in particular, the six-sector obstruction is a proposition proved here, not a conditional hypothesis. The subgroup-embedding argument is also supplied, using the explicitly stated Coxeter exchange/word theorem as standard background. For historical context on Artin parabolic embeddings, see [2].

A metric space is *proper* if every closed bounded set is compact. Our CAT(0) convention is the geodesic triangle-comparison inequality with Euclidean comparison triangles. An action

is *geometric* if it is isometric, proper and cocompact on a nonempty proper space. Action properness means that $\{g : gK \cap K \neq \varnothing\}$ is finite for every compact K; cocompactness means that translates of some compact set cover the space. There is no dimension restriction.

Definition 1.1. Let

$$S = \{a, b, c, x, y, d, t_4, t_5, t_6, t_7, t_8, t_9\}.$$

Define a symmetric Artin matrix $M = (m_{uv})_{u,v \in S}$ as follows. Set $m_{uu} = 1$ for every $u \in S$. On each of the three ordered lists

$$\begin{aligned} \mathcal{B}_1 &= (a, b, x, t_4, t_5, t_6, t_7, t_8, t_9), \\ \mathcal{B}_2 &= (c, b, x, t_4, t_5, t_6, t_7, t_8, t_9), \\ \mathcal{B}_3 &= (a, c, y, t_4, t_5, t_6, t_7, t_8, t_9), \end{aligned} \tag{1}$$

prescribe the standard B_{10} labels: entries in consecutive positions have label 3, and entries in nonconsecutive positions have label 2. Additionally, set

$$m_{da} = m_{db} = m_{dc} = 2.$$

Every remaining off-diagonal label is ∞. Let

$$G := A_M$$

be the resulting Artin group, using the elements of S as its standard generators.

Thus label 2 means $uv = vu$, label 3 means $uvu = vuv$, and label ∞ imposes no relation between the indicated generators. The prescriptions in (1) agree on every overlap. In particular,

$$m_{ab} = m_{bc} = m_{ca} = 3, \qquad m_{at_4} = m_{bt_4} = m_{ct_4} = 2.$$

For an additional check on the specification, the complete list of unordered pairs with label ∞ is

$$\{b, y\},\ \{x, y\},\ \{d, x\},\ \{d, y\}, \quad \{d, t_j\} \quad (4 \leq j \leq 9).$$

There are precisely $6 + 6 = 12$ standard generators.

Theorem 1.2. *The Artin group G of Definition 1.1 admits no geometric action on a nonempty proper* CAT(0) *space.*

2 Stable lengths and centralizer characters

The arguments in this section apply to an arbitrary group G acting geometrically on a nonempty proper CAT(0) space X. Here geometric means that the action is isometric, proper and cocompact. Action properness means that $\{g \in G : gK \cap K \neq \varnothing\}$ is finite for every compact $K \subset X$.

Lemma 2.1 (Elementary comparison inequalities)**.** *Geodesics in a* CAT(0) *space are unique. If $\alpha, \beta : [0, 1] \to X$ are geodesic segments parametrized proportionally to length, then*

$$d(\alpha(t), \beta(t)) \leq (1 - t)d(\alpha(0), \beta(0)) + td(\alpha(1), \beta(1)). \tag{2}$$

In particular, geodesics depend continuously on their endpoints. If m is the midpoint of $[v, w]$, then

$$d(z, m)^2 \leq \frac{d(z, v)^2 + d(z, w)^2}{2} - \frac{d(v, w)^2}{4}. \tag{3}$$

For any four points p, q, r, s,

$$d(p, r)^2 + d(q, s)^2 \leq d(p, q)^2 + d(q, r)^2 + d(r, s)^2 + d(s, p)^2. \tag{4}$$

Proof. The Euclidean median identity, followed by triangle comparison, gives (3) for the midpoint of any chosen geodesic from v to w. Apply this inequality with z the midpoint of a second such geodesic. Its right side is zero, so the two midpoints coincide. Repeating on the two halves and using continuity proves uniqueness.

Comparison for two segments with a common initial endpoint gives $d(\alpha(t), \beta(t)) \le t\, d(\alpha(1), \beta(1))$; reversing the segments gives the corresponding assertion for a common final endpoint. Insert the segment from $\alpha(0)$ to $\beta(1)$ between arbitrary α and β and apply the triangle inequality. This proves (2) and continuous dependence on the endpoints. Finally, let m be the midpoint of $[p, r]$. The triangle inequality gives

$$d(q, s)^2 \le 2d(q, m)^2 + 2d(s, m)^2.$$

Apply (3) to the two terms on the right to obtain (4). □

For an isometry g and a point $o \in X$, define its *stable length* by

$$\ell(g) = \lim_{n\to\infty} \frac{d(o, g^n o)}{n}. \tag{5}$$

To see that the limit exists, set $a_n = d(o, g^n o)$. This sequence is subadditive. For fixed $k \ge 1$ and $n = qk + r$, $0 \le r < k$, one has $a_n \le qa_k + a_r$, hence $\limsup a_n/n \le a_k/k$. Taking the infimum in k proves that the limit exists and equals $\inf_{k\ge1} a_k/k$. Changing o changes a_n by at most twice the distance between the old and new basepoints. Consequently stable length is independent of o, is invariant under conjugation, and satisfies

$$\ell(g^m) = |m|\ell(g) \qquad (m \in \mathbb{Z}). \tag{6}$$

For positive m, this follows by taking a subsequence in (5); for negative m, also use $d(o, g^{-n}o) = d(o, g^n o)$.

Lemma 2.2 (Attainment and positivity)**.** *For a geometric action on a nonempty proper* CAT(0) *space,*

$$\ell(g) = \min_{x\in X} d(x, gx) \qquad (g \in G).$$

If this number is positive, g preserves a geodesic line on which it acts as translation by $\ell(g)$. Moreover, $\ell(g) = 0$ if and only if g has finite order.

Proof. Put $D = \inf_{x\in X} d(x, gx)$ and choose x_n with $d(x_n, gx_n) \to D$. Choose compact K with $GK = X$ and write $x_n = h_n p_n$, $p_n \in K$. For all sufficiently large n, the element $q_n = h_n^{-1} g h_n$ moves p_n by at most $D + 1$. The closed $(D + 1)$-neighborhood K' of K is compact, since X is proper. Both p_n and $q_n p_n$ belong to K', so action properness implies that only finitely many q_n occur. After passing to a subsequence, $q_n = q$ is constant and $p_n \to p \in K$. Then $d(p, qp) = D$. A fixed corresponding conjugator h_n transfers p to a minimizer x for g.

Suppose $D > 0$, and let m be the midpoint of $[x, gx]$. Then gm is the midpoint of $[gx, g^2x]$. Triangle comparison gives

$$D \le d(m, gm) \le \tfrac{1}{2} d(x, g^2x) \le D,$$

so $d(x, g^2x) = 2D$. We prove inductively that the concatenation of $[x, gx], \dots, [g^{i-1}x, g^i x]$ is geodesic. If it is geodesic up to $g^i x$, consider the triangle with vertices $x, g^i x, g^{i+1}x$. The points $g^{i-1}x$ and $g^{i+1}x$ are each at distance D from $g^i x$, on its two incident sides, and are at mutual distance $2D$. Their Euclidean comparison points can have distance at least $2D$ only when the comparison angle at $g^i x$ is π. Thus $d(x, g^{i+1}x) = (i + 1)D$, proving the induction.

Translate these concatenations by all powers of g to obtain a geodesic line. In particular $\ell(g) = D$.

If $D = 0$, a minimizer is fixed by g, and $\ell(g) = 0$. The stabilizer of this point is finite by action properness, so g has finite order. Conversely a finite-order element has zero stable length directly from (5). □

Lemma 2.3 (Quadratic lengths on commuting families)**.** *For pairwise commuting isometries $g_1, \dots, g_k$ of a* CAT(0) *space, write $g^v = g_1^{v_1} \cdots g_k^{v_k}$ for $v \in \mathbb{Z}^k$. The function $v \mapsto \ell(g^v)$ extends uniquely to a continuous seminorm on $\mathbb{R}^k$ whose square is a positive semidefinite quadratic form. For commuting g, h, define*

$$\langle g, h \rangle = \frac{\ell(gh)^2 - \ell(g)^2 - \ell(h)^2}{2}. \tag{7}$$

On any commuting family these pairings are the associated bilinear form; in particular $\langle g^i, h^j \rangle = ij\langle g, h \rangle$. They are invariant under simultaneous conjugation.

Proof. For $v \in \mathbb{Z}^k$ put $\nu(v) = \ell(g^v)$. Commutation, the triangle inequality and (6) imply

$$\nu(u+v) \le \nu(u) + \nu(v), \qquad \nu(mv) = |m|\nu(v), \qquad |\nu(u) - \nu(v)| \le \sum_{i=1}^{k} |u_i - v_i| \ell(g_i).$$

Defining $\nu(v/q) = \nu(v)/q$ for positive integers q gives a well-defined seminorm on $\mathbb{Q}^k$. The displayed Lipschitz bound extends it uniquely and continuously to $\mathbb{R}^k$.

Apply (4) to

$$o, \quad g^{nu}o, \quad g^{n(u+v)}o, \quad g^{nv}o$$

in that cyclic order. Divide by n^2 and pass to the limit. With $Q = \nu^2$, the result is $Q(u+v) + Q(u-v) \le 2Q(u) + 2Q(v)$. Initially this holds for integral u, v; homogeneity and continuity extend it to all real vectors. Apply the same inequality to $u+v, u-v$ and use $Q(2u) = 4Q(u)$ to obtain the reverse inequality. Hence

$$Q(u+v) + Q(u-v) = 2Q(u) + 2Q(v). \tag{8}$$

Set $B(u, v) = (Q(u+v) - Q(u-v))/4$. It is symmetric and continuous, and (8) gives

$$B(u+w, v) + B(u-w, v) = 2B(u, v).$$

Taking $w = u$ gives $B(2u, v) = 2B(u, v)$. Taking $u = (a+b)/2$, $w = (a-b)/2$ now proves additivity in the first variable. Integer and rational homogeneity follow, and continuity gives real homogeneity. Symmetry proves bilinearity. Since $B(u, u) = Q(u) \ge 0$, the form is positive semidefinite. Expanding $Q(u+v)$ gives (7). Conjugacy invariance follows from the corresponding property of stable length. □

Lemma 2.4 (Centralizer characters)**.** *Suppose G acts geometrically on a nonempty proper* CAT(0) *space. For every $g \in G$ the map*

$$\mathrm{C}_G(g) \longrightarrow \mathbb{R}, \qquad h \longmapsto \langle g, h \rangle$$

is a group homomorphism. Thus $\langle g, h_1 h_2 \rangle = \langle g, h_1 \rangle + \langle g, h_2 \rangle$ whenever h_1, h_2 commute with g; they need not commute with one another.

Proof. If $\ell(g) = 0$, Lemma 2.3 gives, for all $t \in \mathbb{Z}$,

$$0 \le \ell(g^t h)^2 = 2t\langle g, h\rangle + \ell(h)^2.$$

Letting t tend to both signs of infinity shows that the pairing is zero. Now suppose $D = \ell(g) > 0$, and choose a point x on the axis from Lemma 2.2. Define

$$b(z) = \lim_{n\to\infty} \bigl(d(z, g^n x) - nD\bigr).$$

The terms decrease by the triangle inequality and are bounded below by $-d(z, x)$, so the limit exists. The function b is 1-Lipschitz.

Fix $h \in \mathrm{C}_G(g)$ and $z, w \in X$. Put $p_n = g^n x$ and $q_n = h^{-1} g^n x$. Since h commutes with g, $d(p_n, q_n) = d(x, h^{-1}x)$ is constant. Apply (4) in two orders to obtain

$$\bigl|d(z, p_n)^2 + d(w, q_n)^2 - d(w, p_n)^2 - d(z, q_n)^2\bigr| \le d(z, w)^2 + d(p_n, q_n)^2.$$

Here $d(z, p_n) = nD + b(z) + o(1)$ and $d(z, q_n) = d(hz, p_n) = nD + b(hz) + o(1)$, with analogous expressions for w. Divide by $2nD$ and let n tend to infinity. The result is

$$b(hz) - b(hw) = b(z) - b(w). \tag{9}$$

Thus $\chi(h) = b(x) - b(hx)$ is a homomorphism on $\mathrm{C}_G(g)$: apply (9) with $z = kx, w = x$ to compute $\chi(hk) = \chi(h) + \chi(k)$. Also $b(x) = 0$, $b(gx) = -D$, and $\chi(g) = D$. Since b is 1-Lipschitz,

$$n|\chi(h)| = |b(x) - b(h^n x)| \le d(x, h^n x), \qquad \text{hence } |\chi(h)| \le \ell(h).$$

Apply this last bound to $g^t h$, for every integer t. Lemma 2.3 gives

$$(tD + \chi(h))^2 \le t^2 D^2 + 2t\langle g, h\rangle + \ell(h)^2.$$

After cancelling the quadratic terms, arbitrarily large positive and negative t force $D\chi(h) = \langle g, h\rangle$. The pairing is therefore a multiple of a homomorphism. □

3 A ten-strand braid inequality

The braid relations on $s_1, \dots, s_{n-1}$ are

$$s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}, \qquad s_i s_j = s_j s_i \quad \text{if } |i - j| \ge 2.$$

In this section the s_i may be elements of any group satisfying these relations; an injective braid subgroup is not needed. Define

$$L_1 = 1, \qquad L_j = s_{j-1} L_{j-1} s_{j-1}, \qquad Z_r = L_1 \cdots L_r, \qquad N_r = \binom{r}{2}. \tag{10}$$

Lemma 3.1 (Braid identities)**.** *For $i \le j - 2$, L_j commutes with s_i. The elements L_j commute pairwise, and Z_r commutes with every s_i for $i < r$. Moreover,*

$$Z_2 = s_1^2, \qquad Z_3 = (s_1 s_2)^3,$$

and $\delta = s_1 \cdots s_{n-1}$ satisfies

$$\delta s_i \delta^{-1} = s_{i+1} \qquad (1 \le i \le n - 2). \tag{11}$$

In a geometric action of the ambient group,

$$\langle s_i^2, Z_r\rangle = \frac{\ell(Z_r)^2}{N_r} \qquad (2 \le r \le n,\ 1 \le i < r). \tag{12}$$

Proof. Induct on j. For $i \le j-3$, both s_{j-1} and L_{j-1} commute with s_i. For $i = j-2$, put $a = s_{j-2}$, $b = s_{j-1}$ and $C = L_{j-2}$. Then $bC = Cb$ and $L_j = baCab$, so

$$aL_j = abaCab = babCab = baCbab = baCaba = L_j a.$$

This includes $j = 3$, when $C = 1$; the assertion for $j = 2$ is empty. Since L_i is a word in $s_1, \dots, s_{i-1}$, pairwise commutation of the L_j follows.

Induct on r to prove centrality of Z_r in the initial block. Both Z_{r-1} and L_r commute with s_i for $i \le r-2$. For the remaining generator $s = s_{r-1}$, write $L = L_{r-1}$, so $L_r = sLs$. The established commutation of L and L_r says $LsLs = sLsL$. Consequently

$$s(LL_r) = sLsLs = LsLs^2 = (LL_r)s.$$

The element s also commutes with Z_{r-2}, proving the assertion; $r = 2$ is immediate. Directly,

$$Z_3 = s_1^2 s_2 s_1^2 s_2 = s_1(s_1 s_2 s_1)s_1 s_2 = s_1(s_2 s_1 s_2)s_1 s_2 = (s_1 s_2)^3.$$

To prove (11), move the final s_i in δs_i left past $s_{i+2}, \dots, s_{n-1}$, replace $s_i s_{i+1} s_i$ by $s_{i+1} s_i s_{i+1}$, and move the new s_{i+1} left past $s_1, \dots, s_{i-1}$. The result is $s_{i+1}\delta$.

Finally apply the character $h \mapsto \langle Z_r, h\rangle$ from Lemma 2.4. All s_i, $i < r$, are in its domain, and their adjacent braid relations show that its values on these generators are equal, say to α. The positive word L_j has $2(j-1)$ letters, and Z_r has $2N_r$ letters. Hence

$$\ell(Z_r)^2 = \langle Z_r, Z_r\rangle = 2N_r\alpha, \qquad \langle Z_r, s_i^2\rangle = 2\alpha,$$

which proves (12). □

Lemma 3.2 (Ten strands suffice)**.** *Suppose a group acting geometrically on a nonempty proper* CAT(0) *space contains elements $s_1, \dots, s_9$ satisfying the ten-strand braid relations. Assume s_1 and Z_{10} have infinite order. Then*

$$\frac{\ell((s_1 s_2)^3)}{3\ell(s_1^2)} = \frac{\ell((s_1 s_2)^3)}{3\ell(s_2^2)} > \frac{1}{2}.$$

The infinite-order assumptions hold whenever there is a homomorphism from the ambient group to $\mathbb{Z}$ sending all the s_i to 1.

Proof. Set $g = s_1^2$, $D = \ell(g) > 0$, and

$$R_r = \frac{\ell(Z_r)^2}{N_r^2 D^2} \quad (2 \le r \le 10), \qquad c_0 = \frac{\langle s_1^2, s_3^2\rangle}{D^2}.$$

All the s_i^2 have length D, by (11). We first claim that

$$\langle s_i^2, s_j^2\rangle = c_0 D^2 \qquad (j - i \ge 2). \tag{13}$$

For $i = 1$, all s_j with $j \ge 3$ commute with g. Apply the character of g to each adjacent braid relation among these s_j. Its values on the s_j are equal, proving the claim for $i = 1$. For general i, j, conjugate (s_1, s_{j-i+1}) by δ^{i-1} and use (11) repeatedly. The greatest index to which the shift formula is applied is $j - 1 \le 8$, so every step is valid. Simultaneous conjugation invariance proves (13).

Equation (12) gives

$$\langle g, Z_r\rangle = N_r R_r D^2. \tag{14}$$

Since $Z_3 = gL_3$, the character of g gives $\langle g, L_3\rangle = (3R_3 - 1)D^2$. For $j > 3$,

$$L_j = (s_{j-1} \cdots s_3) L_3 (s_3 \cdots s_{j-1}).$$

Every displayed factor belongs to $\mathrm{C}_G(g)$, and $\langle g, s_i\rangle = c_0 D^2/2$ for $i \geq 3$. Consequently

$$\langle g, L_j\rangle = (3R_3 - 1 + (j-3)c_0)D^2 \qquad (j \geq 3).$$

Here L_3 is kept as a single factor; there is no use of additivity across its individual letters, which need not commute with g. Sum over $Z_r = L_2 \cdots L_r$, including $L_2 = g$, to obtain the recurrence

$$N_r R_r = 1 + (r-2)(3R_3 - 1) + \binom{r-2}{2} c_0 \qquad (3 \leq r \leq 10). \tag{15}$$

Now compare Z_5 with $Z_5' = \delta^5 Z_5 \delta^{-5}$. Repeated use of (11) sends its generators $s_1, \ldots, s_4$ to $s_6, \ldots, s_9$; the largest index used in an individual shift is 8. Every generator of the first block commutes with every generator of the second. Thus Z_5 and Z_5' commute and have equal length. Write their positive words as $Z_5 = u_1 \cdots u_{2N_5}$ and $Z_5' = v_1 \cdots v_{2N_5}$. First expand using the character of Z_5', and then using the character of each u_p. These are legitimate separate expansions because of the cross-block commutations. Equation (13) and bilinearity on commuting pairs give

$$\langle Z_5, Z_5'\rangle = \sum_{p=1}^{2N_5}\sum_{q=1}^{2N_5} \langle u_p, v_q\rangle = (2N_5)^2 \frac{c_0 D^2}{4} = N_5^2 c_0 D^2.$$

It follows that

$$0 \leq \ell\big(Z_5 (Z_5')^{-1}\big)^2 = 2N_5^2 D^2 (R_5 - c_0), \qquad \text{so } R_5 \geq c_0.$$

Apply (15) with $r = 5$ and $r = 10$:

$$10R_5 = 9R_3 - 2 + 3c_0, \qquad 45R_{10} = 24R_3 - 7 + 28c_0.$$

The first equation and $R_5 \geq c_0$ imply $7c_0 \leq 9R_3 - 2$. Since Z_{10} has infinite order, Lemma 2.2 gives $R_{10} > 0$. Therefore

$$0 < 45R_{10} \leq 24R_3 - 7 + 4(9R_3 - 2) = 60R_3 - 15,$$

and $R_3 > 1/4$. Since $Z_3 = (s_1 s_2)^3$, $N_3 = 3$ and $\ell(s_2^2) = D$, taking square roots gives the conclusion. Finally, a homomorphism sending each s_i to 1 sends s_1 to 1 and Z_{10} to $2N_{10} = 90$, so both have infinite order. □

4 A triangular complex and controlled diagrams

Throughout this section let

$$T = \langle a, b, c \mid aba = bab,\ bcb = cbc,\ cac = aca\rangle, \qquad \mathcal{E} = \{(a,b), (b,c), (c,a)\}.$$

All cells below retain their labels and their translate in T. Two cells are not identified merely because they have the same vertex set.

4.1 The presentation and the links

For each $(u, v) \in \mathcal{E}$ introduce two new symbols p_{uv}, h_{uv}. Within this block write $(x_1, x_2, x_3, h) = (u, v, p_{uv}, h_{uv})$, and impose the three triangular relations

$$x_i x_{i+1} = h \qquad (i = 1, 2, 3), \tag{16}$$

where subscripts are read modulo three. This is a presentation of T: the first two relations eliminate $h = x_1 x_2$ and $x_3 = x_2^{-1} x_1 x_2$, after which the third is precisely $x_1 x_2 x_1 = x_2 x_1 x_2$. The converse substitution verifies all three relations from this braid relation. The eliminations are independent because the added symbols are private to their blocks.

Let $\mathcal{A}$ be this set of nine generators. Define a labelled triangular Cayley complex K: its vertices are the elements of T; for every $g \in T$ and $q \in \mathcal{A}$ it has an oriented edge (g, q) from g to gq; and for every relation in (16) and every g it has a triangle with vertices

$$g, \quad gx_i, \quad gh.$$

The exponent homomorphism taking a, b, c to 1 takes each x_i to 1 and each h to 2. Thus the vertices, sides and corners of each of these triangles are distinct.

Lemma 4.1. *The link of every vertex of K has no circuit of length less than six. It has no parallel edges.*

Proof. Write q^+, q^- for the outgoing and incoming directions of a labelled generator q. For a fixed relation $x_i x_{i+1} = h$, the initial, middle and final corner give exactly the link edges

$$\{x_i^+, h^+\}, \qquad \{x_i^-, x_{i+1}^+\}, \qquad \{x_{i+1}^-, h^-\}. \tag{17}$$

At any specified vertex v, these occur respectively in the translates $g = v$, $g = v x_i^{-1}$ and $g = v h^{-1}$. In particular, the translate of the triangle is determined by its link corner. One block of the link is a theta graph, with three internally disjoint paths

$$h^+ - x_1^+ - x_3^- - h^-, \qquad h^+ - x_2^+ - x_1^- - h^-, \qquad h^+ - x_3^+ - x_2^- - h^-.$$

The three blocks intersect only at the signed directions of a, b, c. Their edges are distinct, including the only edges joining two shared directions,

$$\{a^-, b^+\}, \quad \{b^-, c^+\}, \quad \{c^-, a^+\}. \tag{18}$$

This proves the absence of parallel edges.

A simple circuit lying in one block has length at least six. Otherwise, split it into $k \geq 2$ maximal nonempty paths in individual blocks. If $k = 2$, both endpoints of each path lie in the intersection of the same two blocks. These endpoints must be the two signs of the one shared generator; their distance in each block is three. If $k = 3$, the circuit visits all three blocks and changes blocks at a signed direction of each of a, b, c. In a block, equal signs of the two shared generators have distance at least two. Opposite signs have distance at least three, except for the directed minus-to-plus choice in (18), whose distance is one. If all three signs agree, the total length is at least $2 + 2 + 2$. Otherwise the cyclic sequence of three signs has one minus-to-plus and one plus-to-minus transition, giving the lower bound $1 + 3 + 2 = 6$.

Finally suppose $k \geq 4$. A block path of length one must be an edge in (18). Each shared signed direction lies on at most one such edge, so two length-one paths cannot be consecutive on a simple circuit. At most $\lfloor k/2 \rfloor$ paths have length one; the others have length at least two. The circuit length is at least $2k - \lfloor k/2 \rfloor \geq 6$. □

4.2 Zero signed cell counts on spheres

A *combinatorial map* from a triangulated surface to K specifies the identities of all target cells. An edge maps to a specified edge, in either orientation, or to a vertex. A triangle maps by its vertex correspondence homeomorphically to a specified target triangle, or affinely into a specified edge or vertex. The latter triangles are called collapsed. Maps agree on common sides. After orienting the surface and every target triangle, each noncollapsed source triangle has a sign $+1$ or -1 relative to its target orientation.

Lemma 4.2. *For a combinatorial map of a finite oriented triangulated two-sphere to K, the sum of the signs of the occurrences of every individual target triangle is zero.*

Proof. In each noncollapsed triangle draw a node with three legs going to its side midpoints. In an edge-collapsed triangle join the midpoints of its two nonconstant sides; a vertex-collapsed triangle requires no arc. Joining across source edges gives a planar trivalent graph, together with circles containing no nodes. Discard these circles. Each node remembers its target triangle and sign, and each graph edge pairs sides with the same target-edge identity.

Thicken each connected component into node disks and edge ribbons. The group label on either side of a ribbon is constant: crossing an edge-collapsed triangle preserves the endpoint of the target edge. At a node, consecutive ribbon sides have the common group label of the corresponding triangle corner. Every boundary component of the regular neighborhood therefore has one constant vertex label. Cap its boundary circles and take the cell decomposition dual to the graph. Since the graph was embedded in a sphere, the capped neighborhood is a sphere: thickening a spanning tree first gives a disk, and adding each remaining edge as an embedded band divides a complementary disk into two disks. This also proves Euler's formula for the resulting cell decomposition. Its faces are triangular, but it need not be a simplicial complex; incidences are counted with multiplicity.

The labels rule out problematic self-incidences. A ribbon cannot join two legs of the same node, since these legs represent distinct target edges. Distinct corners of one face cannot become the same dual vertex, since their target vertices are distinct. A bridge in the graph would have both ribbon sides on the same boundary circle, contradicting the distinct endpoint labels of its target edge. Multiple edges between different nodes cause no difficulty.

A *fold* consists of two faces joined across the same side of the same target triangle. They have opposite signs. Delete a fold in a small disk around its two nodes and their chosen connecting ribbon. The four remaining legs, called i, j, j, i in cyclic order according to their target sides, can be reconnected in pairs i to i and j to j without crossing. Their endpoint labels agree, including the label at the common target corner. The operation preserves all target-cell signed counts and planarity. If the two nodes have additional connecting edges, choose the disk thin enough that the middle parts of those edges remain outside it. Discard any resulting circles without nodes, and cap the remaining components as before.

Suppose nodes remain but there is no fold. At a dual vertex, successive corners give a closed walk in the link of its target vertex. A backtrack would traverse the very same target corner twice on the two sides of a common edge. Formula (17) then identifies the same translated target triangle, giving a fold. The two faces are distinct by the self-incidence exclusions above. Thus every such link walk is nonbacktracking, and it has length at least six by Lemma 4.1. If F, E, V count the faces, edges and vertices of one component, respectively, then

$$3F = 2E, \qquad 2E \geq 6V, \qquad V - E + F = 2.$$

The first two inequalities imply $V - E + F \leq 0$, a contradiction. Hence a fold can always be deleted while nodes remain. Each deletion removes two nodes, so the process terminates. All signed counts have been preserved, and at the end every one is zero. □

4.3 Uniform approximation, including triangle interiors

A triangulation of a parameter surface means a homeomorphism from a finite triangulated surface to it. Refinements in the next lemma stay within existing faces and have matching boundary subdivisions. In particular, a prescribed subcomplex is preserved as a subset, although the new triangle coordinates need not be affine in the old ones.

Lemma 4.3. *Suppose T acts geometrically on Y and fix $y \in Y$. There is a constant C_1 depending only on this action, basepoint and presentation such that the following holds. For every continuous map $f : \Sigma \to Y$ from a finite triangulated two-sphere, and every prescribed subcomplex $B \subset \Sigma$, there is a refinement preserving B and a combinatorial map $\Sigma \to K$ such that, for every final triangle Q, every one of its group-vertex labels s, and every parameter point $p \in Q$,*

$$d_Y(sy, f(p)) \le C_1. \tag{19}$$

The constant is independent of f and of the required subdivision.

Proof. Choose R for which Ty is R-dense. Properness implies that only finitely many elements move y by at most $2R+1$. Therefore some integer M satisfies

$$d_Y(sy, ty) \le 2R+1 \quad \Longrightarrow \quad d_T(s,t) \le M,$$

where d_T is word distance for a, b, c and their inverses. Put $D = \max\{d_Y(y, ay), d_Y(y, by), d_Y(y, cy)\}$.

First refine the parameter triangulation, respecting B, until every face image has diameter at most one. At each vertex v choose a label s_v with $d_Y(s_v y, f(v)) \le R$. Choose once per unoriented edge a word path of length at most M between its endpoint labels; use its reverse for the opposite orientation. Add three constant steps at each end of each path and subdivide the edge accordingly. A face boundary is now a labelled null loop of length at most $3(M+6)$. After translating a chosen original vertex label to 1, there are only finitely many possible boundary types: the letters come from $\{a^{\pm1}, b^{\pm1}, c^{\pm1}, 1\}$ and their number is bounded.

We explain why each type has a finite triangulated disk filling with this exact parameter boundary and the permitted simplex maps. Reserve three consecutive constant steps at the chosen base vertex. The remaining word is trivial in the triangular presentation, hence is a finite product of conjugates of its relators and their inverses in the free group. Thus it can be reduced to the empty word by free insertions or deletions, triangular-relator insertions or deletions, and constant-step moves, all outside the reserved steps. Inserting a conjugate wrw^{-1} can be performed by first inserting an out-and-back journey ww^{-1} by free moves, then inserting r at its tip. Each elementary move consequently takes place in one target vertex, edge or triangle.

Realize each move between two disjoint parameter circles as an annulus. The unchanged edges are connected by rectangular strips, split into two collapsed triangles. The changed arcs bound a polygon with all vertex images in the one target simplex supporting the move; triangulate this polygon with the corresponding affine maps. If an arc $p_0, \dots, p_k$ is deleted at a vertex q of the other circle, use the fan with triangles $qp_{j-1}p_j$ instead of identifying parameter vertices. The radial edges qp_0 and qp_k are distinct even when their labels coincide. The preceding strip, fan and following strip have one interval as their link at q, so the parameter surface has no pinched point. Reverse this construction for insertion at an empty arc. The three reserved edges supply a nonempty unchanged part and ensure that every intermediate circle has at least three edges.

Stack the finitely many annuli, then cap the final three-step constant circle with a cone mapped to its vertex. This is an ordinary triangulated disk. Identify it with the original face by a boundary-preserving homeomorphism: view the annuli as successive collars in a disk and the last cone as its central disk, and extend the prescribed outer parametrization radially.

These fillings glue along the common, already fixed edge subdivisions, and every new triangle stays in its original face. Thus B remains a subcomplex.

Fix one such filling for each of the finitely many normalized boundary types. Let L be the largest word length of any vertex label in any of them; it is finite. Private generators of the triangular presentation are words in a, b, c, so they are included in this bound. In a translated filling based at an original vertex v_0 labelled s_0, every label s satisfies $d_T(s_0, s) \le L$. For every point p of that original face,

$$\begin{aligned} d_Y(sy, f(p)) &\le d_Y(sy, s_0y) + d_Y(s_0y, f(v_0)) + d_Y(f(v_0), f(p)) \\ &\le DL + R + 1. \end{aligned}$$

Every final triangle lies in that face, so $C_1 = DL + R + 1$ proves the asserted bound at all its points. □

5 The six-sector obstruction

Throughout this section put

$$T = \langle a, b, c \mid aba = bab,\ bcb = cbc,\ cac = aca \rangle, \qquad \mathcal{E} = \{(a, b), (b, c), (c, a)\}, \qquad z_{uv} = (uv)^3.$$

We prove the geometric obstruction needed for the construction. The proof uses the two elementary diagram lemmas, Lemmas 4.2 and 4.3, for the triangular Cayley complex K of T. The space on which T acts below is not assumed to have dimension two.

Proposition 5.1 (Six-sector obstruction)**.** *There is no geometric action of T on a nonempty proper* CAT(0) *space such that, for each $(u, v) \in \mathcal{E}$,*

$$\frac{\ell(z_{uv})}{3\ell(u^2)} = \frac{\ell(z_{uv})}{3\ell(v^2)} > \frac{1}{2}. \tag{20}$$

The denominators are positive in any geometric action: the homomorphism $T \to \mathbb{Z}$ sending a, b, c to 1 shows that their squares have infinite order. Adjacent braid generators are conjugate, so their stable lengths are equal. For the proof, suppose that the action in the proposition exists on Y, and fix $y \in Y$.

5.1 Orbit control and separating coordinates

Write d_T for the word metric associated to a, b, c and their inverses.

Lemma 5.2. *There are constants $R, C < \infty$ such that Ty is R-dense in Y and*

$$d_T(s, t) \le C(1 + d_Y(sy, ty)) \qquad (s, t \in T). \tag{21}$$

Proof. Choose a compact set D with $TD = Y$ and put $R = \max_{p \in D} d_Y(y, p)$. This gives the density assertion. For $L = d_Y(sy, ty)$, divide the geodesic from sy to ty into $N = \max\{1, \lceil L \rceil\} \le L + 1$ pieces, each of length at most one. Choose orbit points within R of the subdivision points, retaining sy and ty at the ends. Consecutive labels have increments q with $d_Y(y, qy) \le 2R + 1$. There are only finitely many such q, by properness of the action and compactness of the closed ball of radius $2R + 1$ about y. Take $C \ge 1$ to bound all their word lengths. Adding the lengths of these increments proves the estimate. □

Take an equilateral Euclidean triangle with side lines D_a, D_b, D_c, and let r_u denote reflection in D_u. Let $\mathcal{L}$ consist of all images of these lines under the group generated by r_a, r_b, r_c. Set

$$E = \bigoplus_{D\in\mathcal{L}} \mathbb{R}e_D, \qquad \left\|\sum_D \xi_D e_D\right\|_1 = \sum_D |\xi_D|.$$

Every vector has finite support. Reflections permute the basis vectors, and therefore act linearly and isometrically on E.

Lemma 5.3. *The formulas $u \cdot \xi = e_{D_u} + r_u\xi$ for $u = a, b, c$ define an affine isometric action of T on E. Its orbit map $V(t) = t \cdot 0$ satisfies*

$$\|V(t) - V(s)\|_1 \le d_T(s, t). \tag{22}$$

For $(u, v) \in \mathcal{E}$, the elements u^2 and z_{uv} act by pure translations, with respective vectors

$$L_u = 2e_{D_u}, \qquad L_{uv} = 2(e_{D_u} + e_{D_v} + e_{D_{uv}}), \tag{23}$$

where D_{uv} is the third mirror through $D_u \cap D_v$. The six lines $D_a, D_b, D_c, D_{ab}, D_{bc}, D_{ca}$ are distinct.

Proof. The two side reflections at a vertex generate reflections in three lines spaced at angles $\pi/3$. The two words uvu and vuv have the same linear part. Their translation parts are, respectively,

$$e_{D_u} + e_{r_u D_v} + e_{r_u r_v D_u} = e_{D_u} + e_{D_{uv}} + e_{D_v}$$

and the same sum in reverse order. Hence every defining braid relation is satisfied. A generator or its inverse moves 0 by a vector of ℓ^1 norm one. The cocycle identity gives $V(t) - V(s) = r_s V(s^{-1}t)$, where r_s is the linear part of the action of s. This proves (22).

Since r_u fixes e_{D_u}, squaring the affine action of u gives L_u. If $\Delta = uvu = vuv$, then $z_{uv} = \Delta^2$. The linear part of Δ is an involution preserving the sum of the three mirror vectors just displayed. Squaring therefore gives L_{uv}. The third mirror at each vertex of the original triangle is parallel to, and different from, its opposite side. The three opposite directions are different, proving the final assertion. □

5.2 The parameter disk and its large-scale images

For each $(u, v) \in \mathcal{E}$, consider both commuting pairs

$$(U, H) = (u^2, z_{uv}) \quad \text{and} \quad (U, H) = (v^2, z_{uv}).$$

Commutation follows directly from the braid relation: for $\Delta = uvu = vuv$, one has $\Delta u = v\Delta$ and $\Delta v = u\Delta$, so $\Delta^2 = z_{uv}$ commutes with both generators. On the exponent plane of each pair, let $\|\cdot\|_{U,H}$ be the stable seminorm of the commuting family. Thus its value at $(i, j) \in \mathbb{Z}^2$ is $\ell(U^i H^j)$, and its square extends to a positive semidefinite quadratic form. The nonnegative quadrant is parametrized by (λ, μ).

Lemma 5.4. *Each of the six stable forms is positive definite, and its quadrant is a Euclidean sector of angle strictly between 0 and $\pi/3$. Gluing along the common rays, in cyclic order*

$$a^2,\ z_{ab},\ b^2,\ z_{bc},\ c^2,\ z_{ca},\ a^2, \tag{24}$$

gives a Euclidean cone of total angle $0 < \Omega < 2\pi$. Its radius-one part P is a topological disk. There is a continuous injective map $v_* : P \to E$ which, in the sector for (u^2, z_{uv}), is

$$v_*(\lambda, \mu) = \lambda L_u + \mu L_{uv}, \tag{25}$$

and has the analogous formula for the sector for (v^2, z_{uv}).

Proof. For integers i, j and positive integers m, the pure translation formulas give $V((U^iH^j)^m) = m(iL_u + jL_{uv})$ in the first sector. Combining the previous two lemmas, dividing by m, and passing to the limit yields

$$\|iL_u + jL_{uv}\|_1 \le C\|(i,j)\|_{U,H}.$$

By rational homogeneity and continuity this holds for all real i, j. The vectors L_u, L_{uv} are linearly independent: the D_{uv} coordinate first determines j, and then the D_u coordinate determines i. Thus the stable quadratic form is positive definite. The proof for the other sector is the same.

The centralizer-character property gives

$$\langle u^2, z_{uv}\rangle = \frac{\ell(z_{uv})^2}{3}.$$

Indeed, the character $h \mapsto \langle z_{uv}, h\rangle$ is defined on u, v. Their braid relation makes its two values equal, say to α; the word $(uv)^3$ gives $\ell(z_{uv})^2 = 6\alpha$, whereas u^2 gives 2α. Consequently the sector angle θ satisfies

$$\cos\theta = \frac{\langle U, H\rangle}{\ell(U)\ell(H)} = \frac{\ell(H)}{3\ell(U)} > \frac{1}{2}.$$

Positive definiteness also gives $\cos\theta < 1$. Therefore $0 < \theta < \pi/3$.

On a shared ray corresponding to W, exponent $t \ge 0$ has distance $t\ell(W)$ from the origin on either side. The gluings are therefore isometric along rays and give the stated disk. We denote its radial coordinate by r; within each sector it is its Euclidean distance from the origin. No assertion that the cone P is CAT(0) is needed.

The formulas for v_* agree on shared rays. To prove injectivity, a nonzero extra-mirror coordinate D_{uv} specifies the pair (u, v) and the value of μ. Subtracting 2μ from the D_u, D_v coordinates then determines the remaining exponent and which of the two sectors contains the point. If both excesses vanish, the point is on their common H-ray. If all extra-mirror coordinates vanish, the point is on a unique squared-generator ray or is the origin. These are exactly the ray identifications already made. This proves injectivity. □

For positive integers n, in each sector set

$$t_n(p) = U^{\lfloor n\lambda\rfloor}H^{\lfloor n\mu\rfloor}.$$

The labels agree along shared rays.

Lemma 5.5. *There are continuous maps $f_n : P \to Y$, with $f_n(0) = y$, and a constant C_0 independent of n and p, such that*

$$d_Y(f_n(p), t_n(p)y) \le C_0. \tag{26}$$

Within each individual sector,

$$\frac{d_Y(f_n(p), f_n(p'))}{n} \longrightarrow \|p - p'\|_{U,H} \tag{27}$$

uniformly in p, p'. Uniformly on P one has

$$\frac{d_Y(y, f_n(p))}{n} \longrightarrow r(p), \qquad \frac{V(t_n(p))}{n} \longrightarrow v_*(p). \tag{28}$$

If A_1, A_2 are disjoint nonempty closed subsets of P, then there is $\delta > 0$ such that, for all sufficiently large n,

$$d_Y(f_n(A_1), f_n(A_2)) \ge \delta n. \tag{29}$$

Proof. Triangulate each full exponent quadrant by squares of side $1/n$, each split along a diagonal. Send its vertex $(i/n, j/n)$ to $U^i H^j y$; send edges to the proportionally parametrized geodesics joining their endpoint images. On each triangle extend by geodesically coning the opposite edge to the remaining vertex. Restrictions to common edges agree. Geodesic convexity bounds all image distances from a vertex by the maximum of the corresponding vertex distances; commutation bounds these vertex distances independently of i, j, n. Restrict to the radius-one sector. Every point is in a grid square with lower-left vertex carrying its label $t_n(p)$, so this proves (26). The constructions agree along the shared rays and glue to a continuous map; the six sectors admit a common constant C_0.

For a fixed commuting pair define $F(i,j) = d_Y(y, U^i H^j y)$ on $\mathbb{Z}^2$. The triangle inequality gives

$$|F(i,j) - F(i',j')| \le d_Y(y, Uy)|i - i'| + d_Y(y, Hy)|j - j'|.$$

For a rational vector $q = (a/m, b/m)$ with $m > 0$, the vector $\lfloor nq \rfloor$ differs by a bounded vector from $\lfloor n/m \rfloor (a, b)$. The definition of stable length therefore gives

$$\frac{F(\lfloor nq \rfloor)}{n} \longrightarrow \frac{\ell(U^a H^b)}{m} = \|q\|_{U,H}.$$

This includes negative a, b. On a bounded real region, use a finite rational net, the displayed Lipschitz estimate, and the $O(1/n)$ rounding error. They upgrade this convergence to uniform convergence on that region. For p, p' in one sector, the difference of their floored exponent vectors differs by a bounded vector from the floor of $n(p' - p)$. Their orbit-label distance is F of that difference. Together with (26), this proves (27). Taking $p' = 0$ proves the radial assertion. The mirror assertion follows directly from the translation formulas; its error is in fact $O(1/n)$.

Finally, compactness of P and injectivity of v_* give

$$\eta = \min_{p \in A_1,\, q \in A_2} \|v_*(p) - v_*(q)\|_1 > 0.$$

For sufficiently large n, the mirror distance between $t_n(p)$ and $t_n(q)$ is at least $\eta n/2$, uniformly for such p, q. The orbit and mirror bounds imply

$$d_Y(f_n(p), f_n(q)) \ge \frac{\eta n}{2C} - 1 - 2C_0.$$

Reducing the positive linear coefficient proves (29). □

5.3 A second filling away from the origin

Lemma 5.6. *There is $\epsilon > 0$ such that, for all sufficiently large n, the loop $f_n|_{\partial P}$ extends continuously to a second disk whose entire image lies outside the open ball $B_Y(y, \epsilon n)$.*

Proof. For two points of ∂P, take a shorter angular arc between them. Its angle φ is at most $\Omega/2 < \pi$. Unfold the sectors along that arc into a Euclidean wedge. The wedge intersected with the unit disk is convex; hence the chord between the endpoints stays inside it, and its length is

$$2\sin(\varphi/2) \le 2\sin(\Omega/4) < 2.$$

The chord breaks into at most seven pieces, each contained in one sector: a sector containing both endpoints might occur in two end pieces, and every other sector occurs at most once. Applying the uniform estimate (27) to these pieces gives, uniformly for $p, p' \in \partial P$,

$$d_Y(f_n(p), f_n(p')) \le 2\sin(\Omega/4)n + o(n).$$

Fix $p_0 \in \partial P$ and cone the boundary loop to $f_n(p_0)$ by geodesics. Continuous dependence of geodesics on endpoints gives a continuous disk map. If q lies on the segment $[A, B]$, the triangle inequality at each endpoint gives

$$d_Y(y, q) \geq \frac{d_Y(y, A) + d_Y(y, B) - d_Y(A, B)}{2}.$$

Here both endpoints are boundary images, so both distances from y are $n + o(n)$ uniformly. The whole coning segment is consequently at distance at least $(1 - \sin(\Omega/4))n - o(n)$ from y. Take $\epsilon = (1 - \sin(\Omega/4))/2 > 0$. □

Glue the second parameter disk to P along their common boundary. The result is a parameter sphere Σ, with a continuous map, again denoted $f_n : \Sigma \to Y$. The parameter sphere can be chosen independently of n; only its map changes.

5.4 Local coordinates and the winding contradiction

Fix one sector, say $(U, H) = (u^2, z_{uv})$. Choose a closed exponent rectangle

$$B = [\lambda_0, \lambda_1] \times [\mu_0, \mu_1]$$

in its interior, with $\lambda_0, \mu_0 > 0$, sufficiently small and close to the origin that

$$0 < r(p) < \min\{\epsilon/3, 1/4\} \qquad (p \in B). \tag{30}$$

Write $\kappa(p) = (\lambda, \mu)$ on B. Choose $p_* \in \operatorname{int} B$ and a smaller closed rectangle $B' \subset \operatorname{int} B$ containing p_* in its interior.

Choose a finite triangulation of Σ for which B is a subcomplex. Here is an explicit justification in the presence of the circular outer boundary of P. Replace each radius-one sector by the triangle whose outer vertices lie on its two boundary rays at radius one. Its outer radial coordinate ρ is at least $\cos(\theta/2) > \sqrt{3}/2$. The radial homeomorphism to the original sector which fixes radii at most $1/2$, and is linear from radius $1/2$ to ρ with terminal value 1, agrees with its neighbors along shared rays. It fixes B. Extend the four sides of B across its parameter triangle to partition that triangle into finitely many convex polygons, and triangulate these polygons and the other sector triangles. Finally cone the subdivided outer boundary to one new parameter vertex for the second disk. This gives the claimed sphere triangulation.

Define the bounded linear map $J : E \to \mathbb{R}^2$ by

$$J\xi = \left(\frac{\xi_{D_u} - \xi_{D_{uv}}}{2}, \frac{\xi_{D_{uv}}}{2} \right).$$

On the chosen sector, (23) gives $Jv_*(p) = \kappa(p)$. No global injectivity of J is required.

Apply Lemma 4.3 to $f_n : \Sigma \to Y$, preserving B as a subcomplex. It supplies a refinement and a combinatorial map of the refined sphere into K. Vertices are labelled by elements $s \in T$, and there is a constant C_1 independent of n such that

$$d_Y(sy, f_n(p)) \leq C_1 \tag{31}$$

for every final parameter triangle, each of its vertex labels s, and every point p of that triangle. A triangle of this map either maps to a specified target triangle or collapses into a specified target edge or vertex. Full target-cell identities, including the group translate, are retained.

Give a vertex labelled s the planar value $JV(s)/n$ and extend affinely over each abstract parameter triangle. This defines a continuous map $q_n : \Sigma \to \mathbb{R}^2$. The abstract affine

coordinates are used even if the homeomorphism placing a triangle in Σ is not affine in the exponent coordinates.

We claim that $q_n|_B \to \kappa$ uniformly. For $p \in B$ and any vertex label s of a final triangle containing p, the all-points estimate and (26) give

$$d_Y(sy, t_n(p)y) \le C_1 + C_0.$$

Lemmas 5.2 and 5.3 then give a constant C_2 such that $\|V(s) - V(t_n(p))\|_1 \le C_2$. The uniform mirror estimate in Lemma 5.5 yields

$$\left\| \frac{V(s)}{n} - v_*(p) \right\|_1 = O(1/n),$$

uniformly in p and all these labels. Apply J. Since $q_n(p)$ is a convex combination of the three vertex values, the same uniform error bound holds between $q_n(p)$ and $\kappa(p)$. This argument is why the estimate at every point of every triangle is needed.

For completeness, the planar winding calculation used next can be described directly. A loop avoiding $w \in \mathbb{R}^2$ has a winding number: choose continuous arguments of its vectors from w on successive short intervals and add their changes, dividing by 2π. The result is an integer and is invariant under a homotopy avoiding w, since the angle changes vary continuously. If an oriented triangulated disk maps affinely into the plane and w avoids all its projected edges, adding the winding numbers of its oriented triangle boundaries cancels all interior edges. Thus their sum is the winding number of the outer boundary. An affine triangle contributes zero when w is outside its convex hull, and contributes its orientation sign when w is inside its nondegenerate image. The latter assertion follows by following the rays from w to the triangle boundary, whose directions make one complete turn. A degenerate triangle contributes zero: its image is a segment or point covered by its projected sides, which avoid w.

Choose a small open planar ball W about $\kappa(p_*)$ whose closure lies in $\kappa(\operatorname{int} B')$. Shrink it if necessary. Uniform convergence to κ implies that, for all sufficiently large n and every $w \in W$, the straight-line homotopy from $q_n|_{\partial B}$ to $\kappa|_{\partial B}$ avoids w, and

$$p \in B,\ q_n(p) = w \quad \Longrightarrow \quad p \in \operatorname{int} B'. \tag{32}$$

Indeed, the distance from $\kappa(p_*)$ to the compact set $\kappa(B \setminus \operatorname{int} B')$ is positive; choose the radius of W and the uniform approximation error with sum smaller than this distance.

For each such n, choose $w \in W$ avoiding every projected edge of the entire finite sphere diagram. This is possible because a finite union of line segments cannot contain an open ball. The winding number of $\kappa(\partial B)$ about w is 1 or -1, according to the orientation. The homotopy and cancellation just described show that the sum of the signed coverings of w by the triangles of B is this same nonzero number. Every contributing triangle is noncollapsed and has a nondegenerate planar image. By (32), it contains a point of B' mapping to w.

We next show that every occurrence anywhere on Σ of a target triangle contributing to this winding count lies entirely in $\operatorname{int} B$. Lemma 5.5, applied to the disjoint compact subsets B' and $P \setminus \operatorname{int} B$, separates their f_n-images by a positive multiple of n. The second disk has image at distance at least ϵn from y, whereas (30) and the radial estimate put $f_n(B')$ at distance at most $(\epsilon/3)n + o(n)$ from y. Consequently the distance between $f_n(B')$ and $f_n(\Sigma \setminus \operatorname{int} B)$ tends to infinity uniformly.

Let τ be a contributing target triangle, and choose an occurrence containing $p \in B'$ with $q_n(p) = w$. Fix one of the three group vertex labels s of τ. Every other occurrence of this same target cell has the same vertex labels. If such an occurrence met $\Sigma \setminus \operatorname{int} B$ at p', then (31) would imply

$$d_Y(f_n(p), f_n(p')) \le d_Y(f_n(p), sy) + d_Y(sy, f_n(p')) \le 2C_1,$$

contradicting that separation for large n. The argument also excludes any occurrence touching ∂B.

To finish, orient each target triangle of K once and for all. Its three group labels have fixed planar values $JV(s)/n$, so its projected affine image is fixed. For a contributing cell τ, let $\eta_\tau \in \{1, -1\}$ be the orientation sign of that planar image relative to the chosen target orientation. The covering sign of each occurrence is η_τ times its signed occurrence as a map to τ. All occurrences of each contributing target cell are in B, by the preceding paragraph. Grouping the nonzero winding sum on B by target cells therefore expresses it as a sum of terms

$$\eta_\tau(\text{total signed occurrences of } \tau \text{ on } \Sigma).$$

Every such term is zero by Lemma 4.2. Distinct target cells are kept separate, even if their planar images coincide. This contradicts the winding number 1 or -1, and proves Proposition 5.1.

6 Parabolic embeddings and amalgam normal forms

We give a word-state proof of the embedding needed below. The only Coxeter-theoretic background used in this proof is the exchange/word theorem: in the Coxeter group

$$W = \langle S \mid s^2 = 1,\ (st)^{m_{st}} = 1 \text{ when } m_{st} < \infty \rangle,$$

the length of an element is its shortest word length in S; multiplying a reduced word by a generator and decreasing its length deletes one letter of that word. The word theorem further says that reduced expressions are related by braid moves. In particular, the subgroup W_I generated by $I \subseteq S$ has the restricted Coxeter presentation, and every reduced expression of an element of W_I uses only letters of I. These are statements about Coxeter groups, not an assumption of Artin parabolic injectivity.

Lemma 6.1 (Coxeter coset facts)**.** *Every left coset $W_I w$ has a unique element r of minimum length. It is characterized by*

$$|ur| = |u| + |r| \qquad (u \in W_I).$$

If r is such an element and $s \in S$, then exactly one of the following alternatives holds:

1. *rs is again a minimum left-coset representative;*
2. *$rs = tr$ for a unique $t \in I$.*

If $J \subseteq S$ generates a finite Coxeter group and r has minimum length in a double coset $W_I r W_J$, put

$$K = \{s \in J : rsr^{-1} \in I\}.$$

Then

$$r^{-1} W_I r \cap W_J = W_K.$$

The minimum representatives of left W_I-cosets in that double coset are precisely rv, where v ranges over the minimum representatives of left W_K-cosets in W_J.

Proof. Here and below, minimum means minimum word length. We recall the exchange arguments to make the specific uses of the Coxeter background explicit. If r is minimum in $W_I r$, successively prepending the letters of a reduced I-word for u to a reduced word for r gives a reduced concatenation: otherwise apply exchange at the first failure. Deleting a letter in the I-word contradicts its being reduced; deleting a letter in the word for r produces a

shorter element of $W_I r$. This proves length additivity. It also proves uniqueness, because $r' = ur$ with both r, r' minimum implies $|r'| = |u| + |r|$, whence $u = 1$.

Moreover, minimum representatives are exactly those with no left descent in I. One direction is immediate. Conversely, write any nonminimum element as ur with r minimum. Length additivity shows that the first letter of a reduced nonempty I-word for u is a left descent.

If $|rs| < |r|$, then rs is minimum: a left descent $t \in I$ of rs would give $|tr| \le |trs| + 1 < |r|$, a contradiction. If $|rs| > |r|$ and rs is not minimum, it has a left descent $t \in I$. Apply exchange to t and a reduced expression for rs. The deleted letter cannot be in the expression for r, since cancelling the final s would then make tr shorter than r. It is therefore the final s, giving $trs = r$. This also proves uniqueness of t.

A minimum element r of $W_I r W_J$ is minimum both on the left and on the right, so length is additive in ur and rv for $u \in W_I$ and $v \in W_J$. Suppose $hr = rv$ with $h \in W_I$, $v \in W_J$. We show by induction on $|h| = |v|$ that $v \in W_K$. For $h \neq 1$, write a reduced expression $h = th'$ with $t \in I$. Exchange applied to $trv = h'r$ deletes a letter in the reduced concatenation rv. It cannot delete from r. It follows that

$$tr = rk \quad \text{for some } k \in W_J.$$

Right length additivity, together with $|tr| = |r| + 1$, gives $|k| = 1$. Thus $k \in J$, and $t = rkr^{-1}$ shows $k \in K$. The equality $h'r = rkv$, with $|kv| = |v| - 1$, completes the induction. The reverse inclusion is immediate from the definition of K.

Every left coset in the double coset has the form $W_I rv$ with $v \in W_J$. Replacing v by its minimum left W_K-coset representative preserves this left W_I-coset. We claim rv is then minimum. If $t \in I$ were a left descent of rv, exchange could again not delete from r. As in the preceding paragraph it would give $tr = rk$ with $k \in K$ and $|kv| < |v|$, contrary to the minimum property of v. Uniqueness of minimum representatives finishes the proof. □

Lemma 6.2 (Parabolic embedding for labels $2, 3, \infty$)**.** *Let A_S have a standard Artin presentation with every finite off-diagonal label equal to 2 or 3. For every $I \subseteq S$, the homomorphism from the Artin group on the restricted matrix on I to A_S, given by the named generators, is injective.*

Proof. Denote the abstract Artin group on I by A_I, and its generator corresponding to $t \in I$ by σ_t. Let D be the set of minimum representatives of the left cosets $W_I \backslash W$. Consider states $(r, h) \in D \times A_I$. For a signed letter σ_s^{ε}, with $\varepsilon \in \{1, -1\}$, make the following update, using Lemma 6.1:

$$(r, h) \longmapsto \begin{cases} (rs, h), & rs \in D, \\ (r, h\sigma_t^{\varepsilon}), & rs = tr \text{ with } t \in I. \end{cases}$$

The update for σ_s^{-1} is the inverse of that for σ_s: in the first case the same state edge is traversed backwards; in the second the same letter is recorded with the opposite sign.

It remains to check the positive defining relations at every state. Fix s, t with $m_{st} \in \{2, 3\}$ and put $J = \{s, t\}$. In the double coset containing the initial state, choose a minimum element r_0. By Lemma 6.1, every state in this double coset is $r_0 v$, where v is a minimum representative for $W_K \backslash W_J$ and $K = \{j \in J : r_0 j r_0^{-1} \in I\}$. The state update therefore reduces exactly to the same algorithm in the finite dihedral group W_J, with any recorded $j \in K$ replaced by $r_0 j r_0^{-1} \in I$. There are just three cases.

If $K = \varnothing$, no letters are recorded. If $K = J$, every letter is recorded, and the two outputs are the defining length m_{st} braid relation on the distinct generators $r_0 s r_0^{-1}, r_0 t r_0^{-1} \in I$. Their Coxeter label is m_{st}, because conjugation preserves the order of their product.

If $|K| = 1$, interchange s, t if necessary and take $K = \{s\}$. For $m_{st} = 2$, the minimum representatives are $1, t$; multiplication by s records s and fixes the representative, while multiplication by t interchanges the representatives without recording anything. Thus st and ts both record exactly one s and end in the same state. For $m_{st} = 3$, the minimum representatives are $1, t, ts$. The transitions are

v	s	t
1	$(1;\, s)$	$(t;\, 1)$
t	$(ts;\, 1)$	$(1;\, 1)$
ts	$(t;\, 1)$	$(ts;\, s)$

where each entry gives (new representative; recorded word). From each of the three initial representatives, both sts and tst record precisely one s and have the same terminal representative. This completes the check of every relation.

Consequently the updates define a right action of A_S on $D \times A_I$. Starting at $(1, 1)$, a word in letters from I keeps its first coordinate equal to 1 and records that exact signed word in the second coordinate. If such a word represents the identity in A_S, its action fixes $(1, 1)$, so its second coordinate is the identity in A_I. This proves injectivity. □

Lemma 6.3 (Normal forms for an amalgam)**.** *Let H_1, H_2 contain an isomorphic copy of a group E. In $H_1 *_E H_2$, both factors embed and their intersection is E. An alternating product of elements of $H_1 \setminus E$ and $H_2 \setminus E$ is nontrivial. If it has at least two syllables, it belongs to neither factor.*

Proof. Choose sets $R_i \subseteq H_i$ of representatives for the cosets hE, including 1 for E. Each element of H_i has a unique expression re with $r \in R_i$ and $e \in E$. Use states

$$(r_1, \ldots, r_n; e),$$

where each r_j is a nonidentity representative, successive representatives come from different factors, and $e \in E$; allow $n = 0$. To right-multiply a state by $h \in H_i$, isolate its last representative if that representative is in R_i, or use an empty representative $r = 1$ otherwise. Rewrite the tail reh uniquely as $r'e'$ in H_i, replace the tail by $r'; e'$, and omit r' if $r' = 1$.

These are invertible operations. Operations from one factor obey its multiplication law, since two successive updates rewrite the same product $rehk$ and the representative decomposition is unique. This remains true if the first update removes the last representative: the second then starts with the empty representative. An element of E merely right-multiplies the final e, regardless of which factor is used. The two factor actions thus induce an action of their presented amalgam.

Every state is reached from the empty state $(\,;1)$ by its displayed word. Every rewriting uses valid relations in the amalgam. If two state words represented the same group element, their actions on $(\,;1)$ would produce the same state. Hence these state expressions are unique normal forms. An element of a factor has at most one representative in its normal form, from that factor, proving the embedding and intersection assertions. Finally, processing an alternating word whose syllables all lie outside E appends one nonidentity representative at every step: the old terminal E-element times the next syllable still lies outside E. Its normal form has exactly the stated number of syllables, proving the remaining assertions. □

7 Recovering the triangle group as a common centralizer

Let H be the Artin group on the induced submatrix indexed by $S \setminus \{d\}$. Lemma 6.2 identifies

$$\langle a, b, c\rangle = T, \qquad \langle a, b, c, t_4\rangle = T \times \langle t_4\rangle \quad \text{inside } H.$$

The direct-product description follows by moving all t_4-letters to the right; the maps to and from the displayed direct product are inverse on generators. In particular, t_4 commutes with T and $t_4 \notin T$. Comparing presentations, and using the embedding of T in H, gives

$$G = H *_T B, \qquad B = T \times \langle d \rangle. \tag{33}$$

Lemma 7.1. *In G one has*

$$\mathrm{C}_G(d) = B, \qquad \mathrm{C}_G(d) \cap \mathrm{C}_G(t_4) = T.$$

Proof. Every element of B commutes with d. Conversely, if $g \notin B$, the normal form from Lemma 6.3 in (33) permits an expression $g = wb$, where $b \in B$ and w is a nonempty alternating product of elements outside T, with its last factor in $H \setminus T$. Since b commutes with d,

$$gdg^{-1} = wdw^{-1}.$$

The right-hand side is reduced as an amalgam word: its middle factor d belongs to $B \setminus T$, with neighboring factors in $H \setminus T$. It lies outside B and therefore cannot equal d. This proves $\mathrm{C}_G(d) = B$.

An element of B has the form td^k, where $t \in T$ and $k \in \mathbb{Z}$. Because t commutes with both d and t_4, this element centralizes t_4 if and only if d^k centralizes t_4. For $k \neq 0$, the word

$$d^k t_4 d^{-k} t_4^{-1}$$

is a reduced alternating word with four syllables in the amalgam, since $d^{\pm k} \in B \setminus T$ and $t_4^{\pm 1} \in H \setminus T$. It is nontrivial. Hence $k = 0$, proving the second assertion. □

We record the geometric fact that turns this algebraic computation into a geometric action. Crucially, the elements being centralized need not commute with one another.

Lemma 7.2 (Simultaneous displacement sublevels)**.** *Suppose a group Γ acts geometrically on a nonempty proper* CAT(0) *space X. Let $q_1, \dots, q_m \in \Gamma$ and set $C = \bigcap_{j=1}^m \mathrm{C}_\Gamma(q_j)$. If $A \geq 0$ is finite and*

$$Y_A = \{p \in X : d_X(p, q_j p) \leq A \text{ for all } j\}$$

is nonempty, then C acts geometrically on the proper CAT(0) *space Y_A, equipped with its inherited metric. Every $g \in C$ has the same stable length in Y_A and in X.*

Proof. Each displacement function is continuous and convex, so Y_A is closed and convex. It is therefore a proper CAT(0) space, and it is C-invariant. Properness of the restricted action follows from properness of the original action.

Choose a compact set $K \subset X$ with $\Gamma K = X$, and let $L = \{p \in X : \mathrm{dist}(p, K) \leq A\}$. The set L is compact, so properness makes $F = \{r \in \Gamma : rL \cap L \neq \varnothing\}$ finite. If hK meets Y_A, choose $p \in K$ with $hp \in Y_A$. Then

$$d_X(p, h^{-1} q_j h p) \leq A, \qquad h^{-1} q_j h \in F \quad (1 \leq j \leq m).$$

Only finitely many conjugate tuples $(h^{-1}q_1h, \dots, h^{-1}q_mh)$ can occur. Choose representatives $h_1, \dots, h_N$. If h and h_i give the same tuple, then $hh_i^{-1} \in C$, so $h \in Ch_i$. It follows that the compact set

$$K_C = Y_A \cap \bigcup_{i=1}^N h_i K$$

satisfies $CK_C = Y_A$. Thus the action is cocompact. Finally, invariance and the inherited metric imply $d_{Y_A}(p, g^n p) = d_X(p, g^n p)$ for $p \in Y_A$ and $g \in C$; division by n proves equality of stable lengths. □

8 The contradiction

Proof of Theorem 1.2. Suppose that G acts geometrically on a nonempty proper CAT(0) space X. Choose $p_0 \in X$ and a finite number

$$A \geq \max\{d_X(p_0, dp_0), d_X(p_0, t_4 p_0)\}.$$

Then

$$Y = \{p \in X : d_X(p, dp) \leq A,\ d_X(p, t_4 p) \leq A\}$$

is nonempty. By Lemmas 7.1 and 7.2, T acts geometrically on Y, with the same stable lengths as in X.

Sending every standard generator of G to 1 defines a homomorphism $\varepsilon : G \to \mathbb{Z}$, since both sides of every Artin relation have the same exponent sum. For each block, the elements s_1 and Z_{10} have images 1 and 90, respectively, and therefore have infinite order. Thus the hypotheses of Lemma 3.2 hold.

Apply Lemma 3.2 to each of the three lists in (1). Their initial pairs are (a, b), (c, b) and (a, c). For a length-three braid pair u, v, one has $(uv)^3 = (vu)^3$, and u, v are conjugate. Consequently the inequalities furnished by these three blocks are exactly

$$\frac{\ell_X((uv)^3)}{3\ell_X(u^2)} = \frac{\ell_X((uv)^3)}{3\ell_X(v^2)} > \frac{1}{2} \quad \text{for } (u, v) = (a, b), (b, c), (c, a).$$

They hold with ℓ_Y in place of ℓ_X, because these stable lengths agree on T. This contradicts Proposition 5.1. □

Remark 8.1. The construction has twelve *standard Artin generators*. It does not establish a minimum number of abstract generators for G, nor optimality among Artin groups with the stated obstruction. If one keeps the manuscript's three disjoint braid attachments and two additional centralizer generators, Lemma 3.2 alone reduces the count to $3 + 3(10 - 3) + 2 = 26$.

References

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